Lecture 12: Gradient; Directional Derivatives; tangent plane

$\frac{dw}{dt} = w_x \frac{dx}{dt} + w_y \frac{dy}{dt} + w_z \frac{dz}{dt} = \nabla w \cdot \frac{d\vec{x}}{dt}$ $\quad \nabla w = \langle w_x, w_y, w_z \rangle$ Gradient of w at some point (x,y,z) $\quad \frac{d\vec{x}}{dt} = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle$

Theorem: $\nabla w \perp$ level surface $\{w = \text{constant}\}$

Example 1: $w = a_1 x + a_2 y + a_3 z$ $\quad \nabla w = \langle a_1, a_2, a_3 \rangle$ ($a_1 = \frac{\partial w}{\partial x}$) $\quad$ level surface: $a_1 x + a_2 y + a_3 z = c$ $\quad$ plane with normal $\langle a_1, a_2, a_3 \rangle$

Example 2: $w = x^2 + y^2$ $\quad \nabla w = \langle 2x, 2y \rangle$ $\quad w = c$ $\quad$ level curves are circles

Proof: Take a curve $\vec{r} = \vec{r}(t)$ that stays on the level c. $\quad$ Velocity $\vec{v} = \frac{d\vec{r}}{dt}$ is tangent to the level $w = c$. $\quad$ By chain rule: $\frac{dw}{dt} = \nabla w \cdot \frac{d\vec{r}}{dt} = \nabla w \cdot \vec{v} = 0$ since $w(t) = c =$ constant

So $\nabla w \perp \vec{v}$. This is true for any motion on $w = c$. $\vec{v}$ can be any vector tangent to $w = c$. $\quad$ Given any vector $\vec{v}$ tangent to level curve/plane, $\nabla w \perp \vec{v}$.

Example: Find the tangent plane to surface $x^2 + y^2 - z^2 = 4$, at $(2,1,1)$? $\quad$ Level set $w = 4$ with $w = x^2 + y^2 - z^2$ $\quad \nabla w = \langle 2x, 2y, -2z \rangle = \langle 4, 2, -2 \rangle$ (normal vector) so eq. is $4x + 2y - 2z = 8$

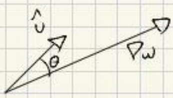
Another way: $dw = 2x\,dx + 2y\,dy - 2z\,dz$; at $(2,1,1) = 4\,dx + 2\,dy - 2\,dz$ $\quad \Delta w \approx 4\Delta x + 2\Delta y - 2\Delta z$ $\quad$ So: level: $\Delta w = 0$ its tangent plane $4\Delta x + 2\Delta y - 2\Delta z = 0$
$= 4(x-2) + 2(y-1) - 2(z-1) = 0$

Directional derivatives: $w = w(x,y) \rightsquigarrow$ know $\frac{\partial w}{\partial x}, \frac{\partial w}{\partial y}$ derivatives in direction of $\hat{\imath}$ and $\hat{\jmath}$

What if we move in direction of $\hat{u}$ = unit vector? Let's look at a straight line trajectory $\vec{x}(s)$ (s = arclength: distance along the line), $\frac{d\vec{x}}{ds} = \hat{u}$ $\quad$ What is $\frac{dw}{ds}$? $\quad$ If $\hat{u} = \langle a, b \rangle$ $\begin{cases} x(s) = x_0 + as \\ y(s) = y_0 + bs \end{cases}$ plug into w; $\frac{dw}{ds}$?

Def: $\frac{dw}{ds}\big|_{\hat{u}}$ directional derivative in direction of $\hat{u}$. $\quad \frac{dw}{ds}\big|_{\hat{u}}$ = slope of slice of graph by a vertical plane $\parallel \hat{u}$ $\quad$ Chain Rule implies: $\frac{dw}{ds} = \nabla w \cdot \frac{d\vec{x}}{ds} = \underbrace{\nabla w \cdot \hat{u}}_{\text{component of } \nabla w \text{ in direction of } \hat{u}}$

Example: $\frac{dw}{ds}\big|_{\hat{\imath}} = \nabla w \cdot \hat{\imath} = \frac{\partial w}{\partial x}$



* maximal when $\cos(\theta) = 1 \Rightarrow \theta = 0 \Rightarrow \hat{u} = \text{dir}(\nabla w)$

Geometry: $\frac{dw}{ds}\big|_{\hat{u}} = \nabla w \cdot \hat{u} = |\nabla w|\,\underbrace{|\hat{u}|}_{=1} \cos(\theta)$

So: dir of ∇w = dir of fastest increase of w. $\quad |\nabla w| = \frac{dw}{ds}\big|_{\hat{u}} = \text{dir}(\nabla w)$

* minimal $\frac{dw}{ds}\big|_{\hat{u}}$ when $\cos(\theta) = (-1) \Leftrightarrow \theta = 180^\circ$ $\quad \hat{u}$ is in direction of $(-\nabla w)$

* $\frac{dw}{ds}\big|_{\hat{u}} = 0$? when $\cos\theta = 0$ $\quad \hat{u} \perp \nabla w$ $\quad \theta = 90^\circ \Leftrightarrow \hat{u}$ tangent to level